

Suggested Mark Scheme

Section A

Answer **all** questions.

- 1 Huntington's disease (HD) is an autosomal dominant disorder characterised by motor dysfunction, cognitive decline and psychiatric disturbances, which lead to progressive dementia and death approximately 15 – 20 years after disease onset.

HD is caused by CAG repeats in the *IT15* gene coding for the huntingtin (HTT) protein. Increase in the number of CAG repeats coding for a glutamine-rich domain in HTT protein seems to induce conformational changes in the protein which causes it to aggregate within cells leading to cell death.

The age at which symptoms of HD first appear is linked with the number of CAG repeats. This is shown in Fig. 1.1.

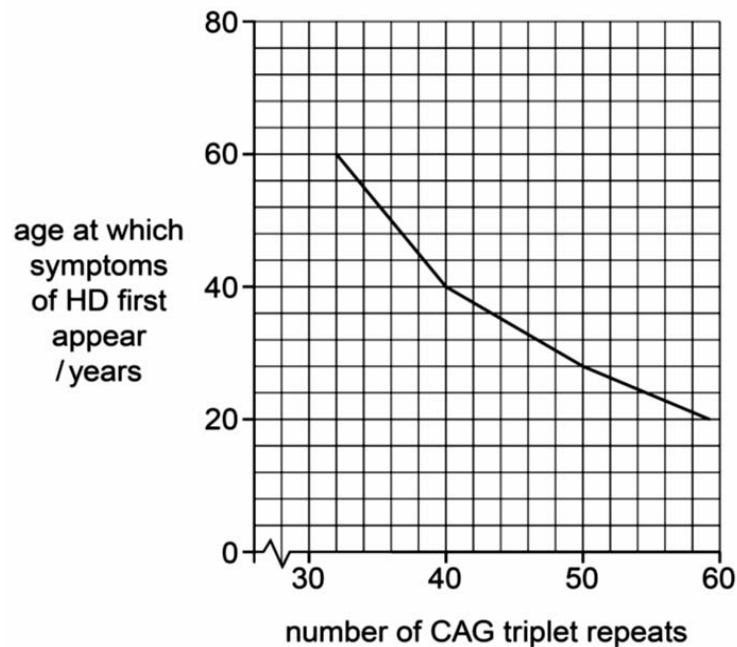


Fig. 1.1

- (a) Describe the relationship between the number CAG repeats and the age at which symptoms of HD first appear.

1. **[trend] larger no. of repeats → earlier age of onset;;**

2. **[QV] 32 repeats → 60 yrs old**
59 repeats → 20 yrs old;;

[2]

Transgenic mice expressing the dominant HD allele develop many of the symptoms of human HD. The progress of the disease in the transgenic mice was monitored by measuring how long the mice could maintain their grip on a rotating rod up to a maximum of 10 minutes (Fig. 1.2). The rod was rotated at 5, 15 or 25 revolutions per minute (rpm).



Fig. 1.2

The performance on the rotating rod of 9- and 12-week old mice from transgenic HD mice was compared to normal control mice. The results are shown in Fig. 1.3.

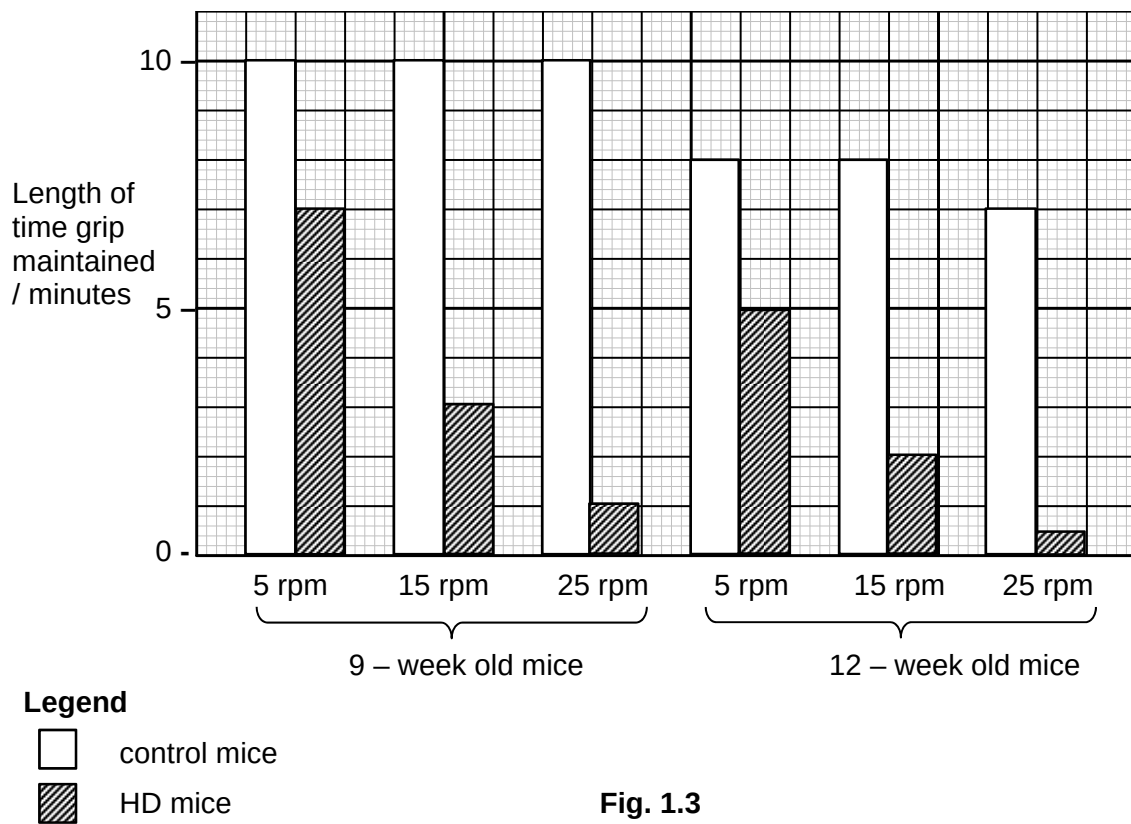


Fig. 1.3

(b) With reference to Fig. 1.3 and information provided, explain the performance of

(i) the two types of 9-week old mice;

1. **control mice maintained grip for 10 min at 5, 15 & 25 rpm**

HD mice maintained grip for shorter periods of time as rpm increases, from 7 min @ 5 rpm to 1 min @ 25 rpm;;

2. **increase in the no. of CAG repeats in HTT prot cause conformational Δ**

which causes it to aggregate within cells $\rightarrow$ death of neurone / brain cells $\rightarrow$ lack of motor / muscle coordination;; [2]

(ii) HD mice at 9 and at 12 weeks old.

1. **12 wk old HD mice maintained grip for shorter periods at all rpms**

time $\downarrow$ from 5 min to $\frac{1}{2}$ min cf to 9 wk old mice holding on for 7 min to 1 min as rpm $\uparrow$ from 5 to 25;;

2. **12 wk old mice had experience longer time since HD onset**

probably more neurone / brain cell death thus less motor / muscle coordination;; OR

12 wk old mice natural degeneration / cell apoptosis of muscle / neurone due to age

as 12 wk old control mice also showed decreased grip time;; [2]

(c) A blood test to detect the HD allele is available for people at risk.

(i) Explain how restriction fragment length polymorphism, agarose gel electrophoresis and nucleic acid hybridisation can be used in the blood test to estimate the age at which HD symptoms first appear for individuals.

1. **cut human DNA with RE to isolate IT15 gene;;**

2. **load DNA fragments from diff individuals into separate wells (in agarose gel near cathode);;**

3. **apply voltage / DC across gel submerged in TBE buffer**

to separate DNA fragments using gel electrophoresis;;

4. **agarose act as molecular sieve (to impede migration of DNA fragments)**

fragments with higher no. of CAG repeats will migrate slower / found nearer to cathode OR

fragments with fewer no. of CAG repeats will migrate faster / found nearer to anode;;

5. **perform Southern blotting using radioactive probe complementary to CAG repeat / HD allele**

subject to autoradiography to visualise position of bands;;

6. **cf position of band to DNA ladder to identify no. of bases in fragment**

cf agst graph in Fig. 1.1 to determine corresponding age of onset;; [6]

- (ii) The location of bands obtained from (b)(i) for different individuals were found to be very close to each other. This made it difficult to determine the size of the DNA fragment. Suggest **one** change to the procedure that can improve the process.

use PAGE for higher resolution;;

[1]

- (d) Explain if HD is likely to be effectively treated using gene therapy.

1. unlikely

as HD is caused by a dominant allele;;

2. introduction of a normal allele does not prevent production of HTT prot thus cell death;;

OR require gene KO / replacement which is difficult to target;;

[2]

[Total: 15]

- 2 Severe Combined Immunodeficiency (SCID) is a group of genetic disorders where patients display severe immunodeficiency and are therefore extremely vulnerable to infectious diseases.

Fig. 2.1 shows the molecular basis of a particular form of SCID where the IL-R2 γ gene is mutated.

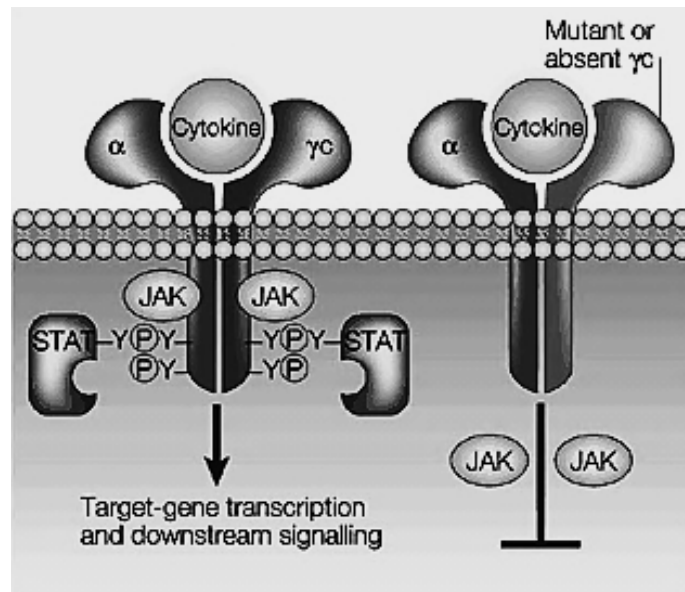


Fig. 2.1

- (a) With reference to Fig. 2.1,

- (i) state the chromosome on which the IL-R2 γ gene is found;

X chr ;

[1]

(ii) describe the molecular basis of the SCID shown.

1. ***IL- R2 γ gene codes for γ c prot / subunit of the IL2R / cytokine receptor***

which binds to cytokine that $\rightarrow$ maturation & dy/dx of lymphocytes ;;

2. ***recessive / LOF mutation $\rightarrow$ leads to incomplete / non-fnal IL2R***

leading to inability to bind cytokine thus no signal transduction that activates Jak & Stat relay prot ;;

3. ***lack of activation of target gene transcription ie no maturation / dy/dx of T and B lymphocytes***

thus leading to reduced nos. $\rightarrow$ severe immunodeficiency ;;

[3]

(b) Early treatment of SCID involved the genetic modification of the patient's own lymphocytes. More recent treatments used genetic modification of the patient's own stem cells. In both cases the viral gene delivery system was used.

(i) Describe **two** problems that may arise when using a viral vector to introduce an allele into a cell.

1. ***Use of viral vector may result in random insertion into genome / insertional mutagenesis***

may lead to LOF of TSG / GOF of proto-oncogene leading to cancer ;;

2. ***Limited size of transgene***

due to limitation in kb of DNA that can fit into capsid ;;

3. ***Can trigger immune response due to viral antigens / surface proteins***

causing rejection and reduction in future transfection efficiency ;;

4. ***Virus may regain virulence / pathogenicity***

virus reactivates virulence factors / pathogenic factors leading to patient infection/inflammatory responses ;;

Any 2 points, AVP

[2]

(ii) Explain why the more recent gene therapy has proved more effective.

1. ***Use of GM stem cells allow sustained pdtn of lymphocytes containing normal therapeutic allele ;;***

2. ***c.f. to limited lifespan of GM lymphocytes ;***

i.e. use of GM stems cells does not require repeated treatments ;;

[2]

- (c) In 2006, scientists in Japan showed that the introduction of four specific genes could reprogramme mature adult cells into a type of pluripotent stem cell called the induced pluripotent stem cell (iPSC). This potentially can replace the usage of embryonic stem cells in gene therapy.

- (i) Describe the characteristics of a stem cell.

1. Undifferentiated / unspecialized cell

as do not possess specialized/tissue-specific cell structures ;;

2. Has potency e.g. totipotent / pluripotent / multipotent

i.e. capable of specializing into different specialized cell types (that performs diff. specialized fns) ;;

3. Have unlimited renewal capacity / capable of long-term self-renewal

due to presence of active telomerase preventing shortening of telomere OR

i.e. can replace worn out / damaged cells ;;

[3]

- (ii) State **one** argument in favour of the use of iPSCs.

1. avoids usage of human embryos (to obtain ESCs)

i.e. avoids the ethical issue of 'sacrificing human lives' ;;

[1]

- (iii) Recently, iPSCs were discovered to retain physiological memory of the adult cell it was derived from. This includes the inability of the cells to synthesise particular proteins coded by its genome.

Suggest why this is so.

1. due to methylation of DNA in differentiated adult cells

i.e. unable to transcribe and translate to synthesise particular proteins ;;

2. AVP

[1]

[Total: 13]

- 3 To boost food production for a rapidly growing global population, crop yields must significantly increase. The C₄ Rice Project aims to improve the photosynthetic capacity of C₃ crops like rice.

Fig. 3.1 shows the production of transgenic C₄ rice plants by the *Agrobacterium*-mediated transformation of rice using calli induced from mature seeds. The genes involved in the C₄ photosynthetic pathways, antibiotic resistance gene and the genes coding for green fluorescent protein (GFP) were added to form the recombinant plasmid in the bacteria.

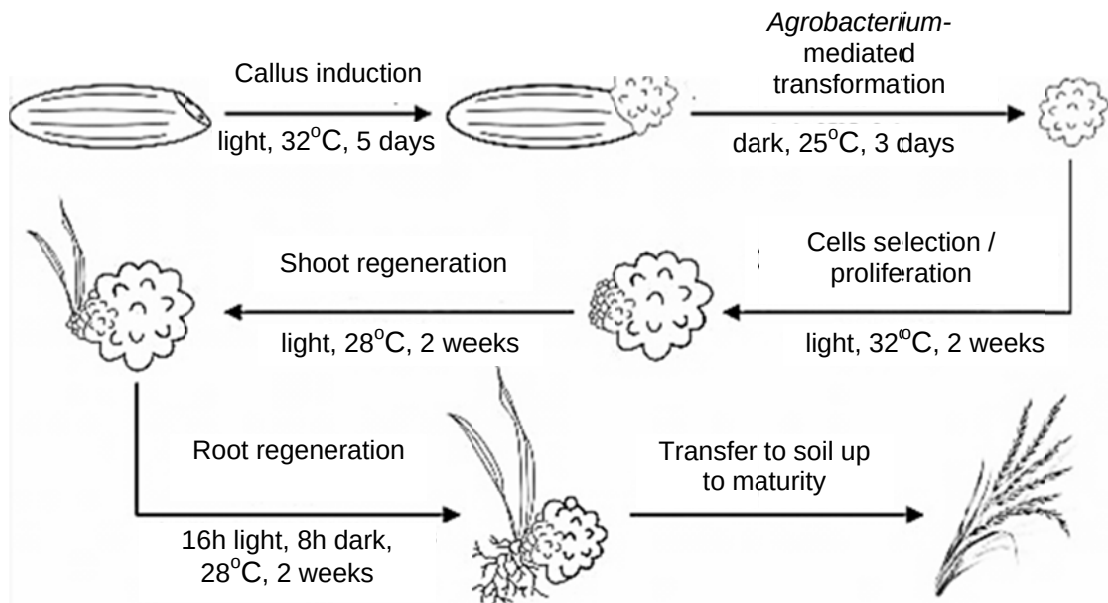


Fig. 3.1

(a) With reference to Fig. 3.1,

- (i) describe the process of callus induction.

1. **rice seeds obtained as explant and sterilised in bleach;;**

2. **place on sterilised nutrient media (autoclaved)**

in aseptic environment / laminar flow hood;;

3. **add intermediate ratios of auxin & cytokinin to growth media**

incubate at 32°C, in light for 5 days;;

® equal conc. of auxin : cytokinin

[3]

- (ii) explain the significance of the GFP in the selection of transformed tissues.
act as selectable marker to allow selection of transformed / recombinant cells

transformed plant cells would fluoresce green;;

[1]

(b) Suggest **one** advantage of using seeds as the explant.

1. **seeds can go into dormancy, would be available at any time of the year i.e. regardless of season;;**
2. **seeds do not have physiological memory, would not mature earlier than expected;;**

(any 1)

[1]

Research suggests that C₄ rice is able to increase the efficiency of utilizing solar radiation that can be translated into yield. Fig. 3.2 shows the net photosynthetic rate of C₃ and C₄ leaves at different light intensity.

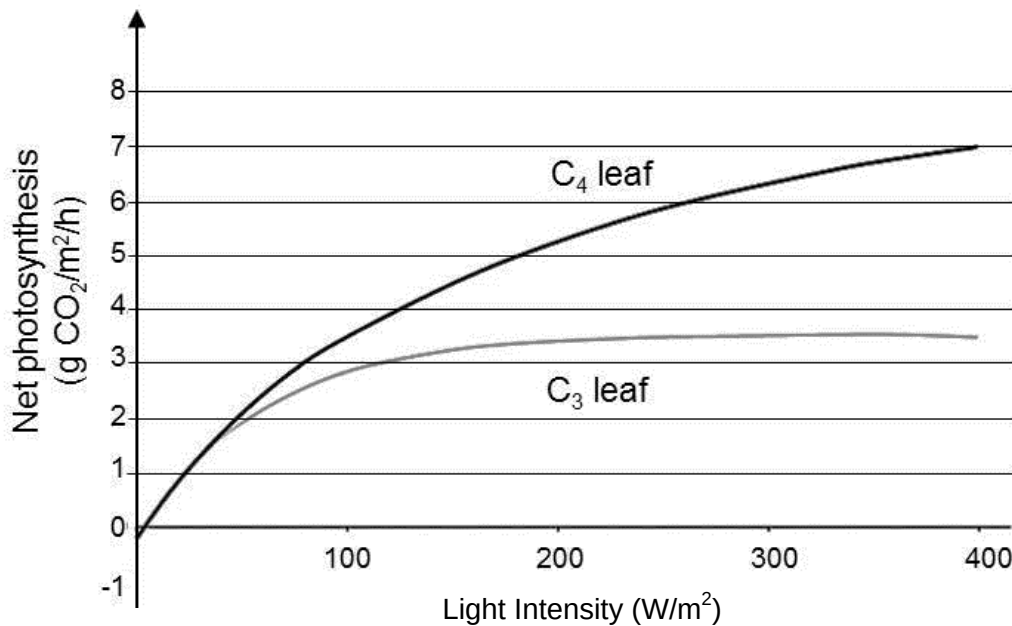


Fig. 3.2

(c) With reference to Fig. 3.2, describe the difference in the net photosynthetic rate in the C₃ and C₄ leaves.

1. **PS rate of C4 higher than C3**

QV (any relevant values);;

2. **C3 reach max PS rate / plateaus at 3.5 g CO₂/m²/h beyond 200 Wm⁻² while PS rate in C4 continues to increase to 7.0 g CO₂/m²/h up to 400 Wm⁻²;;**

3. **LI no longer limiting factor for C3 beyond 200 Wm⁻², other factors e.g. [CO₂] may be limiting**

LI is still the limiting factor for C4;;

[3]

One of the main contributing factor to a higher photosynthetic efficiency in C_4 plants is the absence of a process known as photorespiration. Photorespiration occurs when partial pressure of oxygen is high relative to carbon dioxide. It results in lower carbon dioxide fixation in C_3 plants. This is due to the enzyme Rubisco catalysing a reaction between oxygen and RuBP that results in the production of non-useful by-products. During this process, ATP and reduced NADP are used without leading to the production of sugar.

In C_4 plants, photorespiration is avoided by special arrangement of cells involved in photosynthesis. Additional reactions occur during the light independent stage in mesophyll cells while the Calvin cycle occurs in separate cells known as bundle sheath cells. Fig. 3.3 shows the arrangement of these cells that allows carbon dioxide to be concentrated in the bundle sheath cell.

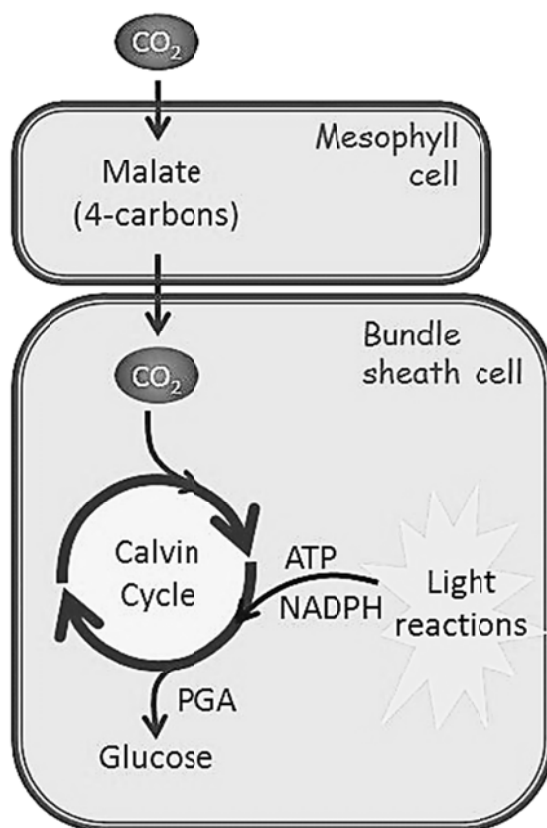


Fig. 3.3

- (d) With reference to Fig. 3.3 and information provided, suggest why C_4 plants are more effective at carbon dioxide fixation at high light intensity.

1. at high LI, there is high P_{O_2} in cells

produced from photolysis of H_2O due to high PS rate ;;

2. light indep stage in C_4 has additional rxns that occurs in mesophyll cells

enz involved in converting $CO_2 \rightarrow$ malate is not affected by high P_{O_2} / does not undergo photoresp ;;

3. CO_2 is released by malate into bundle sheath cells thus P_{CO_2} is high / P_{O_2} is low

Rubisco is not exposed to high O_2 thus preventing photoresp;;

[3]

- (e) Suggest **one** concern of genetically modifying a C_3 plant to a C_4 plant with a higher photosynthetic efficiency.

1. topple ecological balance

gene flow / cross pollination of wildtype relatives $\rightarrow$ superweeds w high PS efficiency;;

[1]

[Total: 12]

4 Planning question

Dormant seeds have a very low rate of respiration. When water is absorbed by dormant seeds, growth hormones are activated. These hormones activate genes that code for enzymes that hydrolyse stored food reserves used in respiration allowing the seed to grow and germinate. The respiration rate can be measured by oxygen usage per unit mass using a respirometer.

You are required to investigate the respiration rate of germinating seeds of different mass.

You are provided with the following materials and apparatus which you must use:

- mung bean seeds of different mass that have been soaked in water for 24 hours
- blue dye
- soda lime pellets
- glass beads
- rubber tubing connected to capillary tube of 4mm diameter
- syringes
- electronic weighing balance

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated
- pipettes, glass rods etc.
- forceps
- ruler
- stopwatch
- thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Theoretical B/grd [2m]

1. Ref to intake of H_2O , activation of growth hormone & enz that hydrolyse food reserves for respiration

ref to aerobic resp for production of ATP for cell division / growth

need for O_2 in resp as final electron acceptor ;;

2. relate measurement of O_2 uptake as indication of resp rate

Ho: mung beans of diff mass may same / diff resp rate with logical suggestion
e.g. same spp thus likely to be the same rate;;

Variables**[Independent variable][1m]**

3. 5 different masses of mung beans of reasonable range & regular intervals

weigh using electronic balance;;

[Dependent variable][1m]

4. respiration rate measured by vol of O_2 uptake

inclusion of soda lime to absorb CO_2

calculated from distance moved by dye droplet;;

[Controlled variables][1m]

5. state 1 variable to be controlled e.g. temperature

description of how it is kept constant ;;

Control setup [1m]

6. description of an appropriate setup using glass beads that occupies similar vol as mung beans

state purpose of control ;;

Procedure [3m]

7. diag. of set up ;;

8. acclimatisation e.g. when droplet starts to move
rxn time e.g. measure droplet move^m over 2 min ;;

9. *logical sequence with appropriate use of apparatus
e.g. forceps to transfer beans, taking up droplet of dye ;;*

Replicates & Repeats [1m]

10. 3 replicates

2 repeats ;;

Results [2m]

11. table of results with appropriate headings, units, calculation of average ;;

12. calculate rate of respiration in mm^3 of O_2 taken in $\text{s}^{-1} \text{g}^{-1}$;;

Safety [1m]

13. soda lime corrosive, skin irritant, wear gloves & eye goggles;;

14. AVP;;

Section B

- 5 (a) Discuss the problems brought about by the human genome project. [6]
1. ***Fairness / right of use of genetic information due to personal genetic info being accessible to multiple parties ;;***
 2. ***Privacy / confidentiality of genetic information due to potential widespread use of individuals' genetic info ;;***
 3. ***Genetic discrimination e.g. knowledge of genetic predispositions may lead to stigmatised by society. not sold insurance etc. ;;***
 4. ***Adequacy of informed consent in decision making due to advancement in genetic technology, more complex and controversial procedures may be used w/o individuals informed consent ;;***
 5. ***Conceptual and philosophical implications, rethinking of human responsibility, free-will, genetic determination, health and diseases, geen doping etc. ;;***
 6. ***Intellectual property rights on ownership of genetic information obtained from HGP ;;***
 7. ***Extent of funding → high cost to seq non-coding sequence with no direct relevance to protein structure and have low priority in time for limited funding for research;;***
 8. ***Resulted in anxiety and frustration as ability to diagnose diseases increased, however no effective treatments proposed;;***

Any 6 points

- (b) Describe the advantages and disadvantages of plant cloning. [6]

Advantages

1. ***Allows rapid multiplication / mass pdtn of plants, giving rise to large number of genetically identical plants as parent /stock plant with desired traits ;;***
2. ***Only a small portion of the plant is needed for the regeneration of whole plants from plant cells / parts that have been genetically modified ;;***
3. ***Allows hybrid production. Desirable traits from sexually incompatible plants may be combined to produce hybrid offspring in protoplast culture ;;***
4. ***Disease free rapidly proliferating meristematic tissue is used as explants for micropropagation to eliminate plant diseases caused by bacteria and viruses ;;***
5. ***Micropropagation can be used for plant species that are difficult to propagate by traditional methods. E.g. plt species that are difficult to pollinate and low seed production ;;***
6. ***Ornamental seasonal plants can be produced continuously, hence meet consumer demand for out-of-season plants ;;***
7. ***large numbers of small containers of plants could be transported more cheaply than adult plants ;;***

Disadvantage

8. **Contamination causing high losses in a period of short time due to bacterial infection ;;**
9. **Plants grown in tissue culture tends to undergo mutation due to exposure to high concentrations of plant growth regulators, hence may lose the chosen trait ;;**
10. **Plant cells may possess physiological memory. Plants generated from non-juvenile cells may mature before they are fully grown ;;**
11. **High cost, need for sophisticated facilities and high labour input, Transplanting plantlets from lab to soil is also labour intensive**
12. **AVP ;;**

Ans must include at least one advantage and one disadvantage

- (c) With reference to solving the demand for food in the world, explain, with an example, the significance of genetically engineered animals to improve their quantity. Include, in your explanation, one concern and one benefit of such GMO. [8]

1. **transgenic salmon with enhanced growth rate ;;**
2. **problem: decreasing harvest by fishing even with farmed salmon also unable to meet demand ;;**
3. **solution: introduce recombinant DNA construct containing salmon growth hormone gene and promoter region of gene coding for antifreeze protein from ocean pout ;;**
4. **as AFP gene promoter is a strong promoter stimulated by cold temp i.e. rearing of GM salmon in cold waters/env ;;**
5. **increases expression of salmon growth hormone which stimulates increased growth rate of salmon ;;**
6. **GM salmon i.e. grows to a larger size at the same age allowing for faster production of salmon to meet consumer demands ;;**
7. **[benefit] shortened growth period reduces risk of disease / feed requirements i.e. lowering costs through fewer fish loss/reduced feed ;;**
8. **[benefit] less pollution from fish farms due to fish excrement and uneaten food ;;**
9. **[concern] ref to threat to ecological balance (e.g. escape from salmon farm and being more aggressive → may displace wild salmon from native habitat) ;;**

10. AVP

Only one benefit and one concern

[Total: 20]